Weather Prediction

Jacob Harmon
Evan Clark

Data Sourcing

Time in Rolla is **Mon 29th Nov 10:33 pm**

Weather » | Hourly » | 3 Hour » | **History »** | Graphs » | Averages » | Widgets » | Weather maps

11/29/2021 Get Weather

Mon 29, Nov 2021

Max: 17°c | Min: 2°c | Sunrise: 07:03 AM | Sunset: 04:48 PM

Moonrise: 01:24 AM | Moonset: 02:05 PM | Phase: Waning Crescent | Illum: 21 %

Time	Weather	Temp	Feels	Wind	Gust	Rain	Humidity	Cloud	Pressure	Vis
00:00		4 °c	3 °c	4 km/h from S	8 km/h	0.0 mm	49%	20%	1023 mb	Excellent
03:00		3 °c	1 °c	8 km/h from S	17 km/h	0.0 mm	53%	15%	1022 mb	Excellent
06:00		2 °c	-1 °c	10 km/h from S	22 km/h	0.0 mm	63%	47%	1021 mb	Excellent
09:00		3 °c	0 °c	12 km/h from S	25 km/h	0.0 mm	70%	56%	1021 mb	Excellent
12:00		13 °c	11 °c	22 km/h from SSW	26 km/h	0.0 mm	44%	3%	1017 mb	Excellent
15:00		17 °c	17 °c	23 km/h from SW	27 km/h	0.0 mm	43%	3%	1013 mb	Excellent
18:00		14 °c	12 °c	16 km/h from SW	26 km/h	0.0 mm	63%	3%	1013 mb	Excellent
21:00		10 °c	8 °c	18 km/h from SW	37 km/h	0.0 mm	71%	5%	1014 mb	Excellent

Data Sourcing

```
▼<div class="card-header">
    <h5>Mon 29, Nov 2021</h5>
  ▼<div class="row">  flex
      <div class="col">Max: 17°c</div> == $0
      <div class="col">Min: 2°c</div>
      <div class="col">Sunrise: 07:03 AM</div>
      <div class="col">Sunset: 04:48 PM</div>
      <div class="w-100"></div>
      <div class="col">Moonrise: 01:24 AM</div>
      <div class="col">Moonset: 02:05 PM</div>
      <div class="col">Phase: Waning Crescent</div>
      <div class="col">Illum: 21 %</div>
    </div>
  </div>
```

```
<div class="col mr-1" style="background-color:#9BF6F3;">4 °c</div>
<div class="col mr-1 d-none d-sm-block" style="background-color:#9BF6F3;">3 °c</div>
<div class="col mr-1 d-none d-sm-block" style="background-color:#9BF6F3;">4 km/h from S</div>
<div class="col mr-1 d-none d-sm-block" style="background-color:#A4F6E9;">8 km/h</div>
<div class="col mr-1" style="background-color:#ffffff;">0.0 mm</div>
<div class="col mr-1 d-none d-sm-block">49%</div>
<div class="col mr-1 d-none d-sm-block" style="background-color:#EEEEEF;">20%</div>
<div class="col mr-1 d-none d-md-block" style="background-color:#C6F56F;">1023 mb</div>
<div class="col d-none d-md-block" style="background-color:#FFFFFF;">Excellent</div>
```

Data Scraping

1.) Set up a dummy browser

```
import selenium.webdriver as webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
```

2.) Browse like a human

```
#Wait for the form to load
elem = WebDriverWait(chrome, 30).until(
    EC.presence_of_element_located((By.ID, "ctl00_MainContentHolder_butShowPastWeather"))
)
#Find the form elements, input and send
chrome.find_element(By.ID, "ctl00_MainContentHolder_txtPastDate").send_keys(date)
chrome.find_element(By.ID, "ctl00_MainContentHolder_butShowPastWeather").click()
```

Data Scraping

3.) Read data

```
from bs4 import BeautifulSoup as BS
```

```
soup = BS(chrome.page_source, "html.parser")
```

4.) Browse for desired elements

```
#Read the first table card
header = soup.find("div", {"class": "card-header"})
subtable = header.find("div", {"class": "row"})
stats = subtable.find_all("div", {"class": "col"})

#High temp
HighTemp = stats[0].text
HighTemp = returnNumb(HighTemp)
```

Data Scraping

```
#Humidity, Wind, Gust, Cloud are all one element type
header = soup.find("div", {"class" : "card-body"})
subtable = header.find("div", {"class": "row text-center"})
data = subtable.find_all("div", {"class" : "col mr-1 d-none d-sm-block"})
avgCloud = 0
avgHumidity = 0
avgWind = 0
avgGust = 0
countLoop = 0
#This is just the only way I could find to easily parse data of the exact same class, by just counting and maintaining an order
for i in range(len(data)):
    if(countLoop == 4):
        avgCloud += returnNumb(data[i].text)
    elif(countLoop == 3):
        avgHumidity += returnNumb(data[i].text)
    elif(countLoop == 2):
        avgGust += returnNumb(data[i].text)
    elif(countLoop == 1):
        avgWind += returnNumb(data[i].text)
    countLoop += 1
    if(countLoop == 5):
        countLoop = 0
# 8 just happens to be the length of the table, I could also have done math to get this
avgWind /= 8
avgGust /= 8
avgCloud /= 8
avgHumidity /= 8
```

Data Scraping

5.) Output with formatting to a data storage

```
dataStr = str(date)+','+str(HighTemp)+','+str(LowTemp)+','+str(avgWind)+','
        +str(avgGust)+','+str(avgHumidity)+','+str(avgCloud)+','+str(avgPressure)+','+str(avgPrecip)+'\n'
print(dataStr)
#Format to a comma separated values text
f.write(dataStr)
```

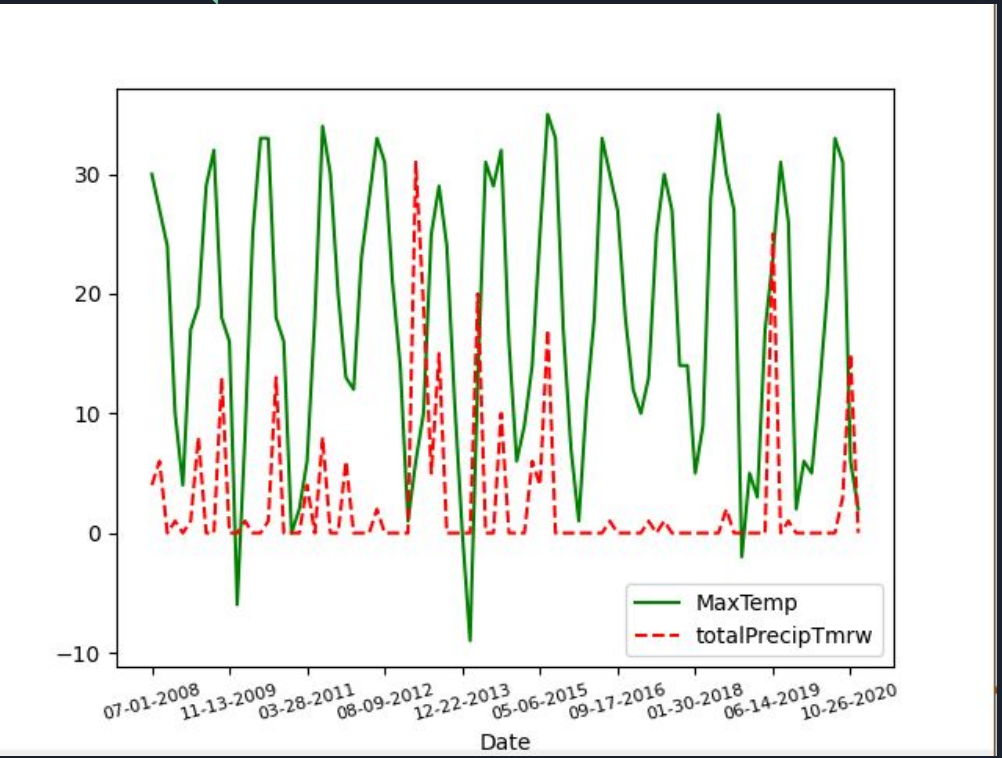
End Result

```
= RESTART: C:\Users\Jacob Harmon\AppData\Local\Programs\Python\Python39\Z5001Mac
hineLearning\DataScraper.py
01-01-2009,6,-4,19.75,30.0,48.125,6.25,1018.375,0.0

01-02-2009,9,-3,9.125,17.5,81.0,7.75,1013.75,0.0

01-03-2009,15,4,19.0,34.625,96.125,81.375,1006.875,0.0

01-04-2009,1,-4,21.25,27.875,73.625,56.25,1022.25,0.0

01-05-2009,2,-4,9.625,13.625,38.625,6.0,1022.625,0.0

01-06-2009,3,-2,9.0,13.75,73.0,45.625,1004.875,0.0

01-07-2009,4,-2,18.375,27.75,70.25,46.5,1002.625,0.0

01-08-2009,3,-6,9.625,15.75,59.25,13.0,1014.75,0.0

01-09-2009,13,0,22.125,39.75,75.0,12.375,1012.5,0.0

01-10-2009,0,-2,21.0,27.625,86.0,99.625,1020.0,0.375

01-11-2009,5,-4,14.875,21.125,73.75,25.0,1023.5,0.0
```

1	Date	MonthYear	MaxTemp	MinTemp	avgWind	avgGust	avgHumidity	avgCloud	avgPressure	avgPrecip
2	Thu 01	Jan 2015	3	-5	16.125	23.125	60.375	56	1025.875	0
3	Fri 02	Jan 2015	6	0	6.25	9.5	84	76.375	1024.125	0.25
4	Sat 03	Jan 2015	5	2	8.375	12.5	95.25	92.75	1013	3.5
5	Sun 04	Jan 2015	-2	-9	23.375	30.75	71.75	59.5	1026.25	0
6	Mon 05	Jan 2015	-3	-10	12.125	16.75	33.5	11.625	1036.875	0

Data Analysis

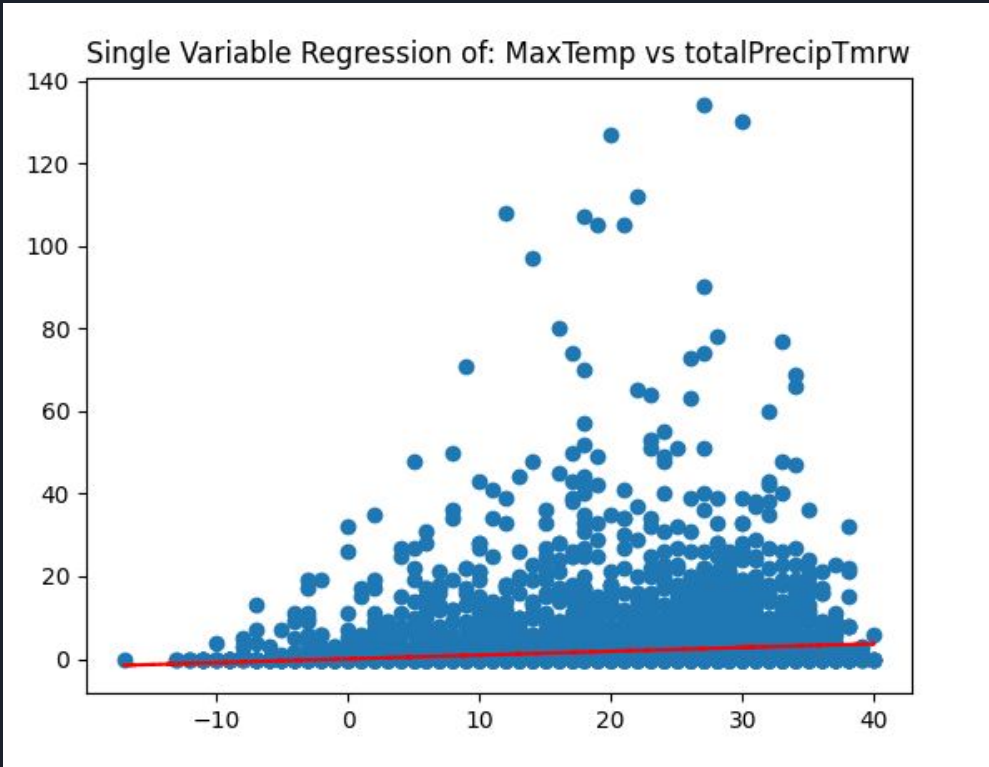
```
MaxTemp correlation with totalPrecipTmrw: 0.045
MinTemp correlation with totalPrecipTmrw: 0.10200000000000001
avgWind correlation with totalPrecipTmrw: 0.127
avgGust correlation with totalPrecipTmrw: 0.14100000000000001
avgHumidity correlation with totalPrecipTmrw: 0.217
avgCloud correlation with totalPrecipTmrw: 0.215
avgPressure correlation with totalPrecipTmrw: -0.111
totalPrecip correlation with totalPrecipTmrw: 0.241
```



$$Correlation = \frac{Cov\,(x, y)}{\sigma x * \sigma y}$$

```
precipTmrw = 0.09173886654153914* MaxTemp + 1.8838590486646745
Cost: 128.7069272500073
R2 Error = 1.061301735536872
Mean Absolute Error: 5.489353555314032

precipTmrw = 0.1412185510274489* MinTemp + 2.3448045139359546
Cost: 127.97506468397235
R2 Error = 1.070931253943937
Mean Absolute Error: 5.4576635530889455

precipTmrw = 0.05914216374102428* avgWind + 2.9165497925621366
Cost: 127.49564539391878
R2 Error = 1.074390959958352
Mean Absolute Error: 5.766352168474598

precipTmrw = 0.06206657947376977* avgGust + 2.4509686326266618
Cost: 126.87776858347846
R2 Error = 1.0786596779957194
Mean Absolute Error: 5.783417123352876

precipTmrw = 0.12213382232761315* avgHumidity + -5.424997965927561
Cost: 126.49468348002988
R2 Error = 1.1059379400458311
Mean Absolute Error: 5.784699596708332

precipTmrw = 0.04958890573627871 4* avgCloud + 1.8543857485429278
Cost: 123.97140388601869
R2 Error = 1.1158417928945625
Mean Absolute Error: 5.777957639784366

precipTmrw = -0.17072557473356048* avgPressure + 177.23680095503266
Cost: 127.35843016827735
R2 Error = 1.075250053667602
Mean Absolute Error: 5.575250311165934

precipTmrw = 0.20187877470608792* totalPrecip + 2.874906852172108
Cost: 123.56803725634148
R2 Error = 1.1165036449974746
Mean Absolute Error: 5.664635907695239
```

Data Analysis



Linear Regression

```
Coefficients:
 [  7.23948964   2.23427132 -19.38643565  24.06279532   0.9391953
   6.17611988   4.78199806  21.37944296]
Mean squared error: 118.99
Coefficient of determination: 0.07
Results:
[0, 1, 0, 3, 7, 1, 0, 0, 27, 24]
Predictions:
[5.84, 5.66, 3.73, 2.97, 4.67, 4.01, 1.59, 1.09, 3.55, 9.85]
Differences from reality:
[  5.8   4.6   3.7  -0.1  -2.4   3.    1.5   1.  -23.5 -14.2]
Average distance from reality:
5.638730853391685
```

Note that heavy rain is classified as anything above 7.6mm per hour.

We are off by 5.63mm a day on average, which isn't a lot over a 24hr period.

Average of 0.7mm/hr we are off, so it's inaccurate, but due to what it's predicting its not awful

Final Equation:

totalPrecipTmrw = 7.23*maxTemp + 2.23*minTemp - 19.38 * avgWind + 24*avgGust

+ 0.94* avgHumidity + 6.17*avgCloud + 4.78 * avgPressure + 21.37 * totalPrecip

Prediction from Linear Regression for 11/30/2021

Normalized values:

MaxTemp	0.596491
MinTemp	0.489362
avgWind	0.339350
avgGust	0.355814
avgHumidity	0.478462
avgCloud	0.187500
avgPressure	0.454988
totalPrecip	0.000000

Prediction:

```
Rain tomorrow: 11.177889026562147
```

Accuracy: Off by:

Anomalies

January 2019 polar vortex caused nearly 400% error

- Maximums and Minimums over late January to early February were around -10C

December 2015 massive flood

- 108mm of rain in one day

What we learned

Even with a large sum of data, weather in localized regions are dependent on data from nearby regions, not just its own data.

What we would change

Predicting an entire days weather is weird and useless, we would like to shift to predicting hourly intervals instead to increase accuracy and usefulness of our predictions.

Try predicting Saint Louis weather based on Rolla weather, or something similar

Extension

Raw data and code available on my website:

Jacobharmon43.github.io

(It is a work in progress still, please go easy on it, also it doesn't work that great on mobile yet)